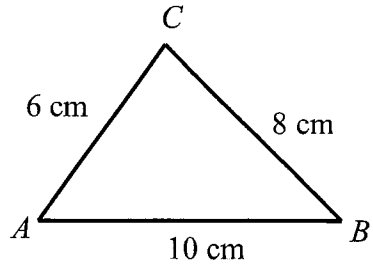
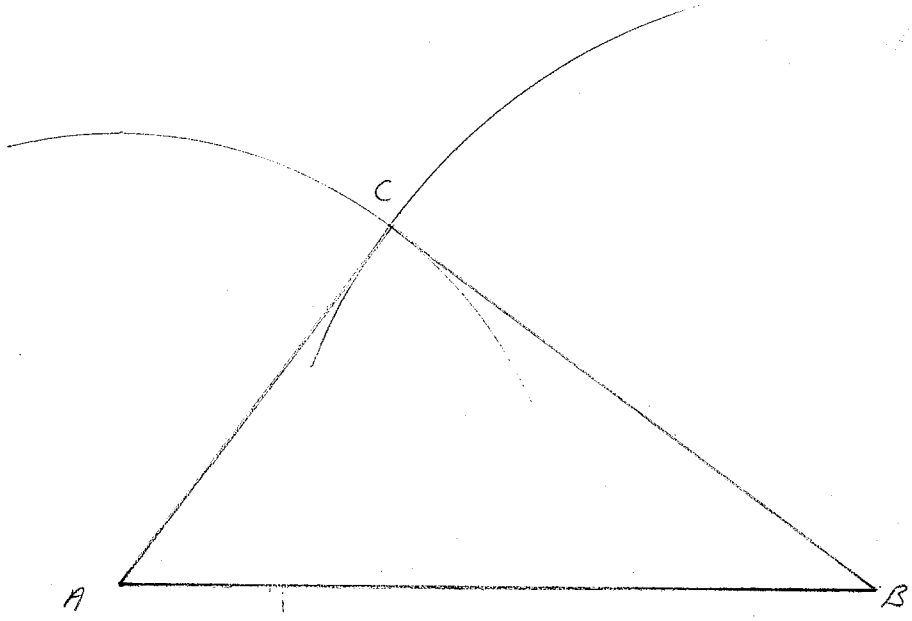
**1** *ABC* is a triangle.




$AB = 10$ cm
$BC = 8$ cm
$AC = 6$ cm

Use ruler and compasses to construct an accurate drawing of triangle *ABC*.

You must show all your construction lines.

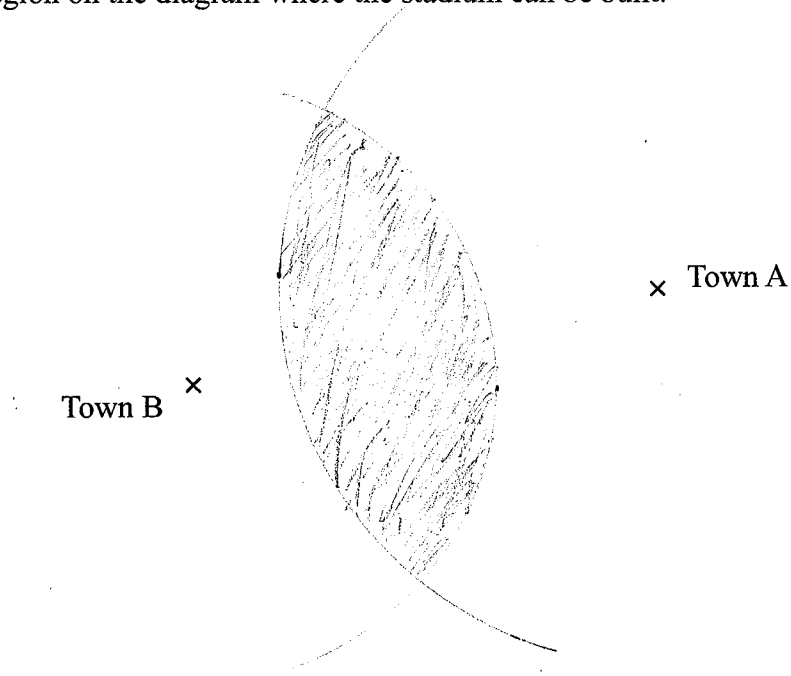



**(Total for Question 1 is 3 marks)**

**2** A stadium is going to be built.
It must be more than 10 km from town A and no more than 8 km from town B.

1 cm represents 2 km 5cm 4cm

Shade the region on the diagram where the stadium can be built.




**(Total for Question 2 is 2 marks)**

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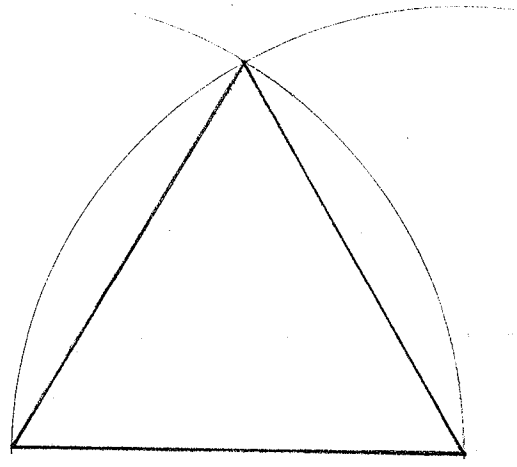
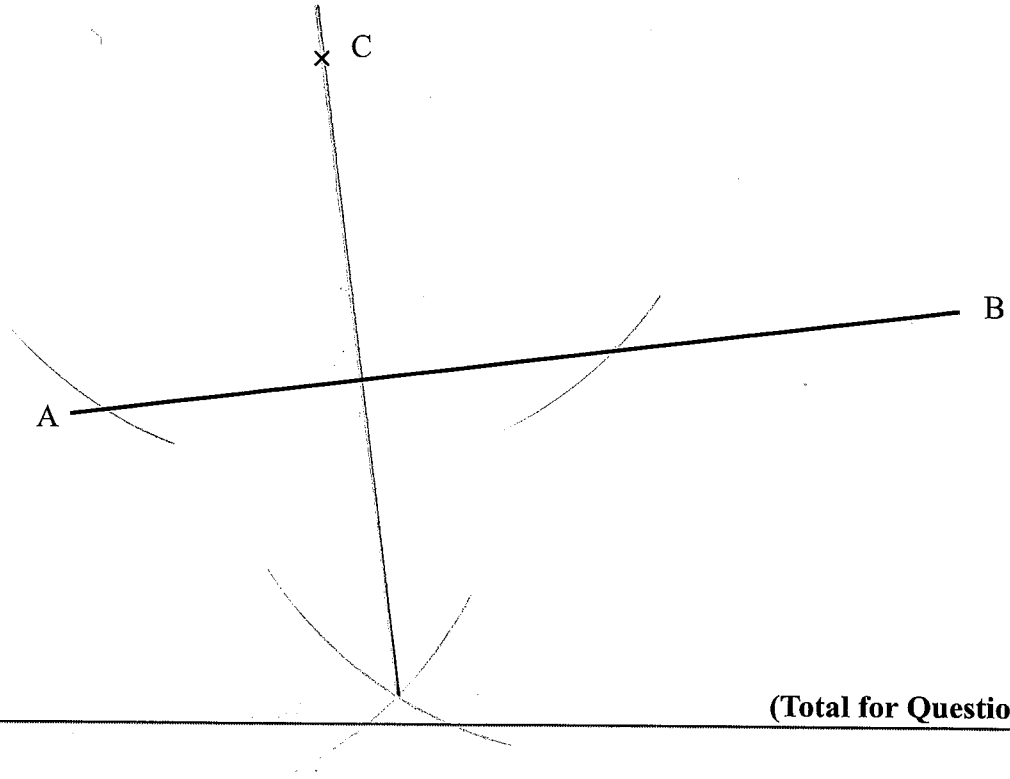
**3** In the space below, use a ruler and compasses to construct an equilateral triangle with side length 6 cm.

You must show all your construction lines.



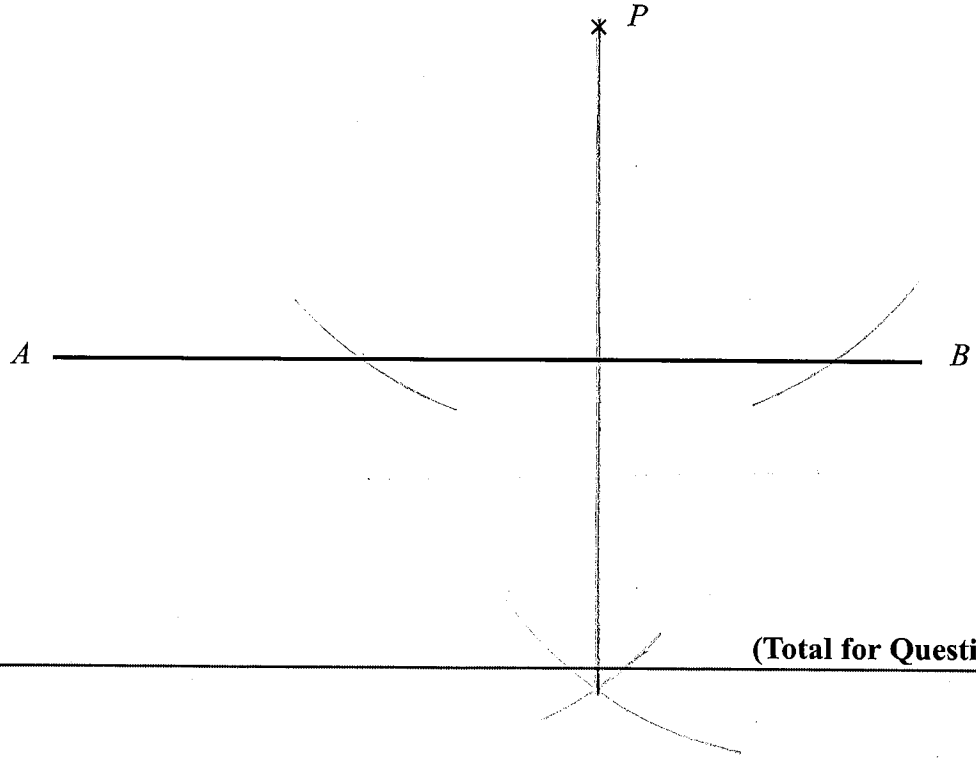
**(Total for Question 3 is 3 marks)**

**4** Use ruler and compasses to construct the perpendicular from point C to the line AB.
You must show all your construction lines.



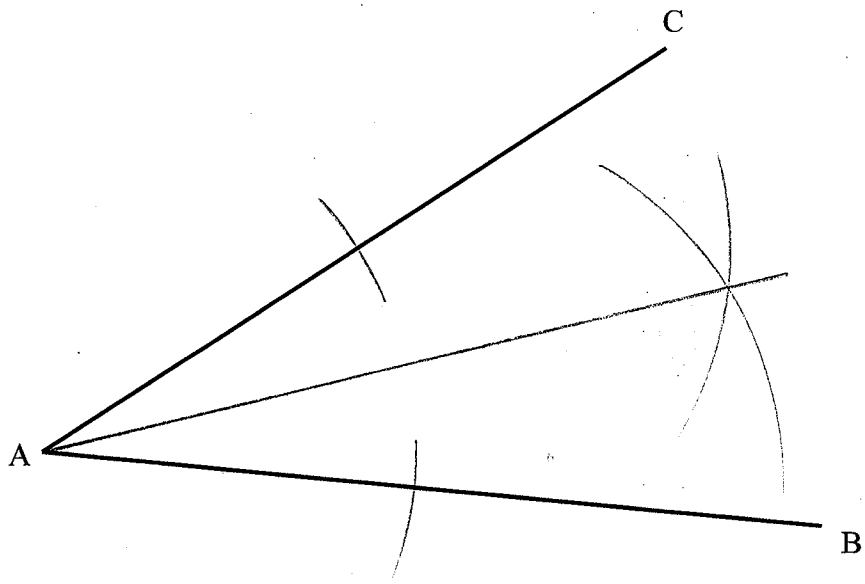

**(Total for Question 4 is 2 marks)**

**5** Use ruler and compasses to construct the perpendicular from point *P* to the line *AB*.
You must show all your construction lines.




**(Total for Question 5 is 2 marks)**

**6** Use ruler and compasses to construct the bisector of angle *BAC*.
You must show all your construction lines.




**(Total for Question 6 is 2 marks)**

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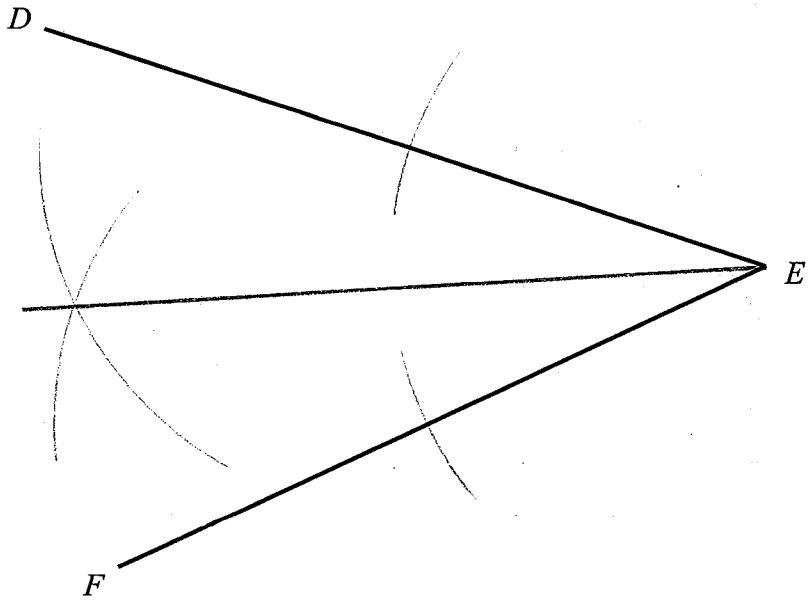
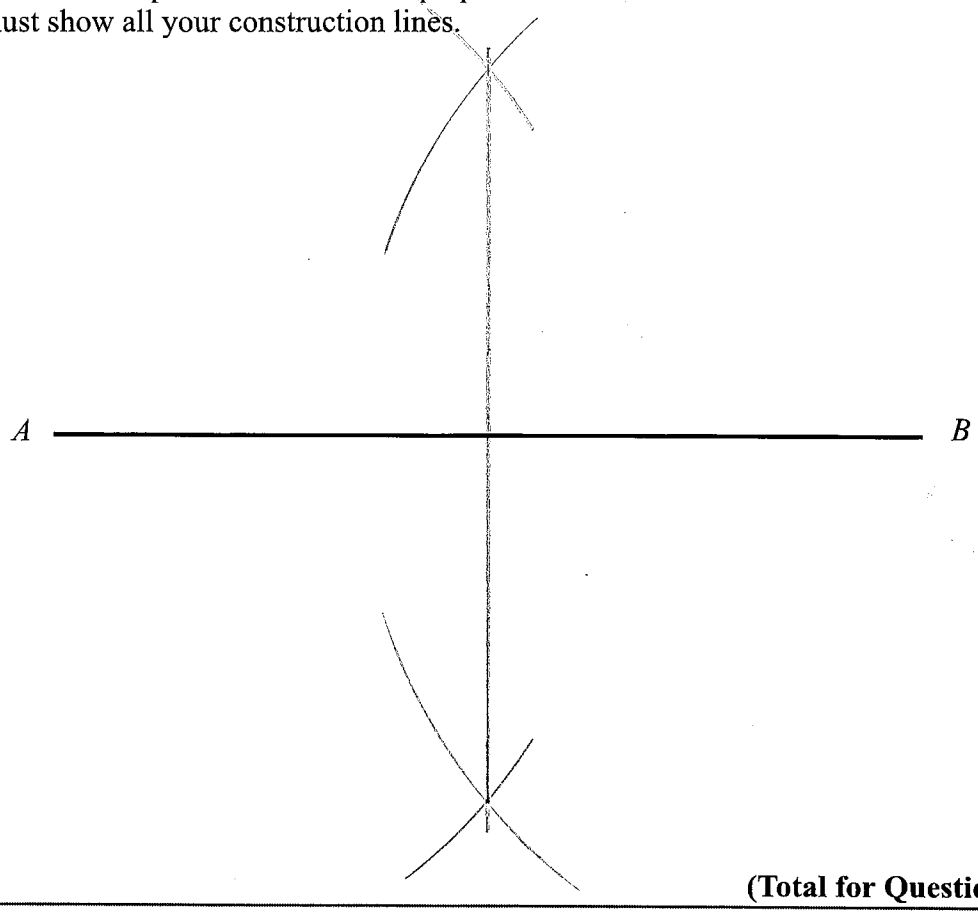
**7** Use ruler and compasses to construct the bisector of angle *DEF*.
You must show all your construction lines.



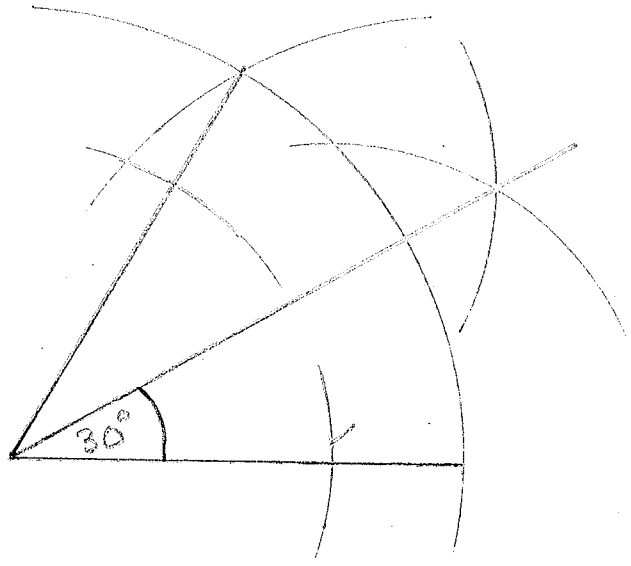

**(Total for Question 7 is 2 marks)**

**8** Use ruler and compasses to construct a perpendicular bisector of the line $AB$.
You must show all your construction lines.




**(Total for Question 8 is 2 marks)**

**9** In the space below, use a ruler and compasses to construct a 30° angle.
You must show all your construction lines.

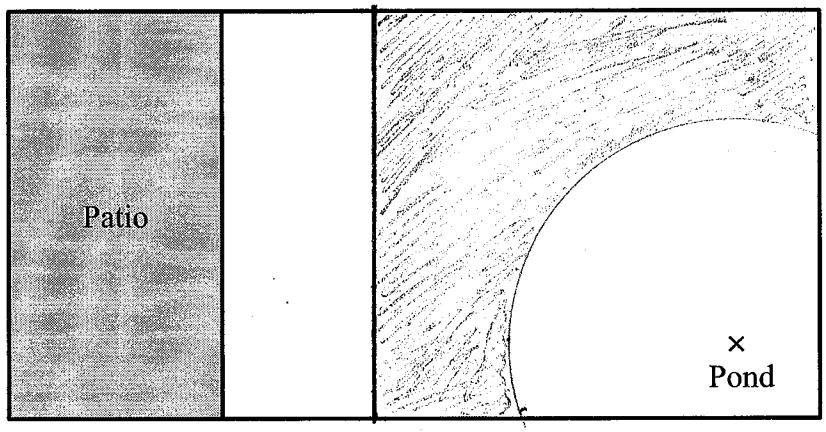



**(Total for Question 9 is 4 marks)**

**10** Here is a scale drawing of a garden.
The scale is 1 cm to 2 m

A tree is going to be planted.
The tree must be more than 4 m from the patio.
The tree must be more than 6 m from the pond.

Shade the region where the tree can be planted.




**(Total for Question 10 is 2 marks)**

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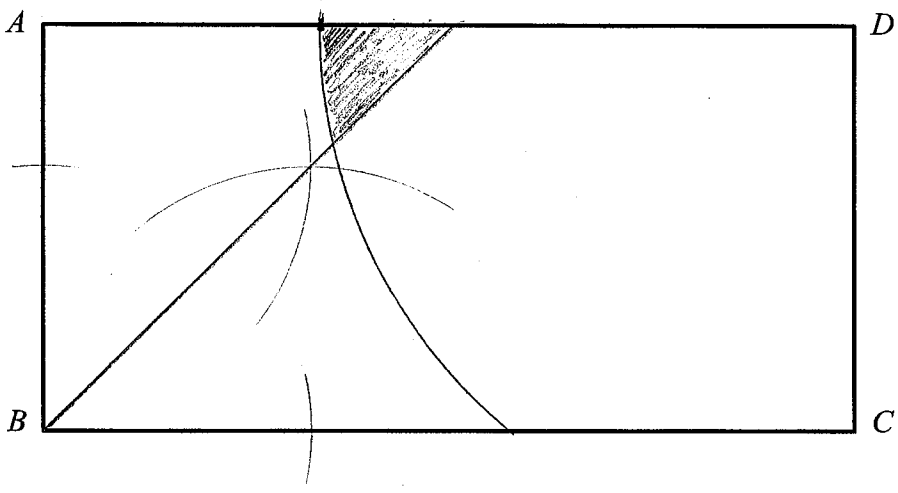
**11** Here is a scale drawing of a room.
The scale is 1 cm to 2 m.

A chair is going to be placed in the room.
The chair must be closer to *AB* than *BC*.
The chair must be less than 14 m from *D*.

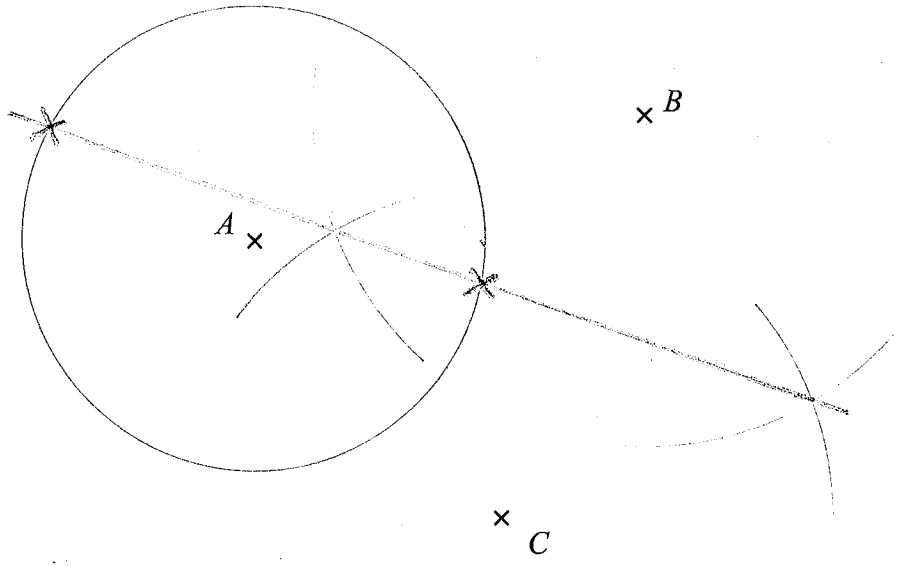
Shade the region where the chair can be placed.




**(Total for Question 11 is 2 marks)**

**12** $A$, $B$ and $C$ are three points on a map.

1cm represents 100 metres.




Point $P$ is 300 metres from $A$.
Point $P$ is equidistant from $B$ and $C$.

On the map, show the possible positions of $P$.

**(Total for Question 12 is 3 marks)**